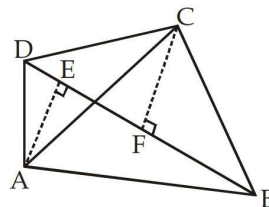


# Quadrilaterals

Quadrilateral is a closed figure in the plane bounded by four sides and consist of four points in which any 3 are non collinear . Parallelogram, Rectangle, Trapezium etc are all types of quadrilaterals.

In the figure, ABCD is a quadrilateral with sides AB, BC, CD, and AD. AC and BD are the diagonals. AE and CF are the perpendiculars on diagonal DB.



## Properties of Quadrilateral

- Sum of all four Interior angle is equal to  $360^\circ$

- Area =  $\frac{1}{2} \times (\text{one diagonal}) \times (\text{sum of perpendiculars to the diagonal from opposite vertexes})$

$$= \frac{1}{2} \times BD \times (AE + CF)$$

- Area =  $\frac{1}{2}$  product of diagonal  $\times$  sine of angle between them.

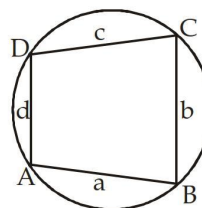
$$= \frac{1}{2} AC \times DB \times \sin \theta$$

- Area of cyclic quadrilateral.

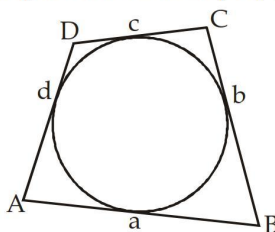
$$= \sqrt{S(S-a)(S-b)(S-c)(S-d)}$$

$$\text{Where, } S = \frac{a+b+c+d}{2} = \text{semi perimeter.}$$

$$\text{and } \angle B + \angle D = \angle A + \angle C = 180$$



- Sum of opposite side of a circumscribed quadrilateral are equal,  $(a+c) = (b+d)$



- Perimeter of quadrilateral is always greater than sum of its diagonals  $(AB + BC + CD + DA) > (AC + BD)$

**Figure formed by joining the mid points of quadrilateral.**

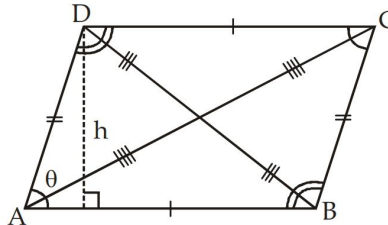
| S. No. | Quadrilateral | Figure formed by joining mid point of Adjacent Side. |
|--------|---------------|------------------------------------------------------|
| 1.     | Parallelogram | Figure formed is Parallelogram                       |
| 2.     | Rectangle     | Figure formed is Rhombus                             |
| 3.     | Square        | Figure formed is Square                              |
| 4.     | Rhombus       | Figure formed is Rectangle                           |
| 5.     | Trapezium     | Figure formed is Parallelogram                       |

so, we can conclude that figure formed by Joining the mid points of any quadrilateral is a parallelogram since Rectangle, Square, Rhombus, Trapezium are all parallelogram.

### Different types of Quadrilateral

**Parallelogram :** Parallelogram or quadrilateral is a figure whose opposite sides are equal and parallel.

$AB = CD, AB \parallel CD, BC = AD, BC \parallel AD$



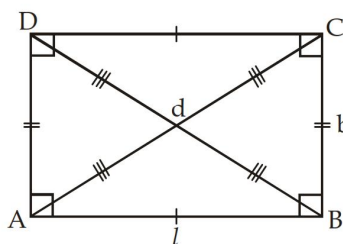
#### Properties:

- Diagonals of parallelogram bisect each other.
- A diagonal of parallelogram divides it into two congruent triangles.  $\Delta ABC$  is congruent to  $\Delta ADC$  and  $\Delta BAD$  is congruent to  $\Delta BCD$ .
- Sum of any two adjacent angles is  $180^\circ$
- Opposite angles are equal
- Diagonals need not bisect at right angle and need not bisect vertex angles
- Parallelogram that is inscribed in a circle is a Rectangle and its diagonal becomes the diameter
- Parallelogram that is circumscribed about a circle is a Rhombus.
- $AC^2 + BD^2 = 2(AB^2 + BC^2)$
- A parallelogram is a rectangle if its diagonals are equal.
- Parallelogram that lie on the same base and between same parallel line are equal in area.
- Area = base  $\times$  perpendicular to base from opposite side.  
 $= AB \times h$
- Area = (Any diagonal)  $\times$  (Sum of perpendiculars to the diagonal from the opposite vertex)
- Area = product of two adjacent sides  $\times$  sine of angle between them  $= AB \times AD \times \sin \theta$
- Area  $= 2\sqrt{s(s-a)(s-b)(s-d)}$ , where a and b are adjacent sides and d is any diagonal.

**Rectangle :** Rectangle is a quadrilateral whose opposite sides are equal and each angle is  $90^\circ$ .

$AB = CD ; BC = AD$

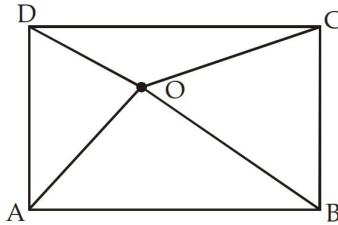
$\angle A = \angle B = \angle C = \angle D = 90^\circ$



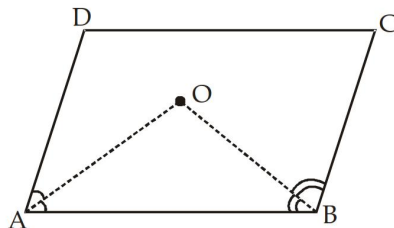
#### Properties:

- Diagonals are equal and bisect each other but need not at right angles.
- Diagonals need not bisect vertex angles.
- For the given perimeters of rectangles a square has maximum area.
- When a rectangle is inscribed in a circle, the diameter of circle is equal to the diagonal of rectangle
- Quadrilateral formed by joining the mid points of intersection of the angle bisectors of a parallelogram is a rectangle.
- Area  $= l \times b = AB \times BC$

- Perimeter =  $2(l + b)$
- Diagonal =  $\sqrt{l^2 + b^2}$ ,  $d = \sqrt{l^2 + b^2}$
- If any point O lies in interior of rectangle then  $OA^2 + OC^2 = OB^2 + OD^2$

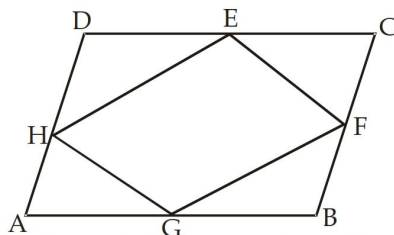


- The bisectors of any two adjacent angle of a parallelogram intersect at right angles  $\angle AOB = 90^\circ$



- Area of parallelogram formed by joining the mid-points of the sides of parallelogram is half the area of original parallelogram

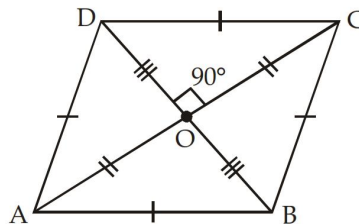
$$\text{area } \parallel \text{ gram (EFGH)} = \frac{1}{2} \text{ area } \parallel \text{ gram (ABCD)}$$



**Rhombus :** A rhombus is a quadrilateral whose all sides are equal and diagonals bisect each other at right angle. Rhombus is a parallelogram whose all sides are equal.

$$AB = BC = CD = AD ; \angle AOB = \angle DOC = \angle AOD = \angle BOC = 90^\circ$$

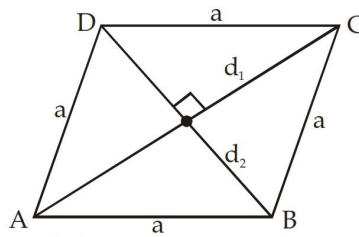
$$AO = CO ; BO = OD$$



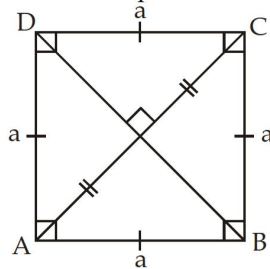
**Properties:**

- Diagonals bisect each other and are at  $90^\circ$  with each other but not necessarily equal in length
- Opposite angles are equal.
- Sum of angles of adjacent vertex equals to  $180^\circ$
- Diagonals bisect the vertex angle
- Area =  $\frac{1}{2} \times \text{product of diagonals} = \frac{1}{2} d_1 d_2 = \frac{1}{2} \times AC \times BD$ .
- Area = product of adjacent sides  $\times$  sine of angle between them.

- Sum of the square of the diagonals is equal to four times the square of side  $d_1^2 + d_2^2 = 4a^2$

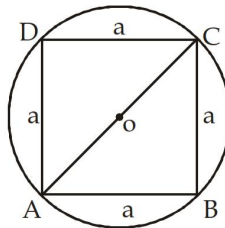


**Square :** A square is a quadrilateral whose all sides are equal and each vertex angle is equal to  $90^\circ$

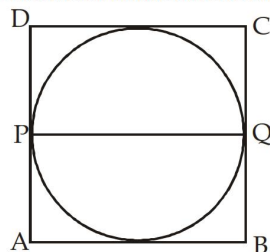


**Properties:**

- All four sides are equal and opposite sides are parallel.
- Diagonals are equal and bisect each other at right angle.
- Diagonal of inscribed square is equal to the diameter of the inscribing circle  
 $AC = \text{diagonal of square} = \text{diameter of circle}$

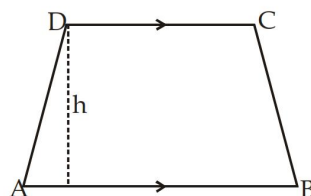


- Side of circumscribed square is equal diameter of the inscribed circle



- Diagonal =  $a\sqrt{2}$
- Area =  $(\text{side})^2 = \frac{(\text{diagonal})^2}{2} = \frac{d^2}{2}$
- Perimeter =  $4a$

**Trapezium :** Quadrilateral whose one pair of opposite sides are parallel and other two sides are not parallel is called as trapezium.

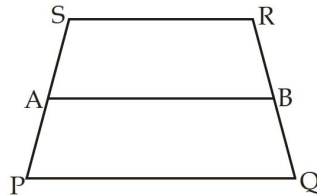




**Properties:**

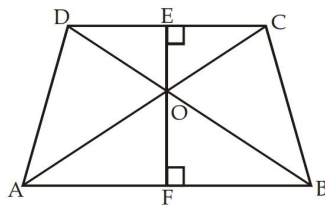
- Line Joining the mid points of non parallel sides of trapezium is equal to half of the sum of parallel sides

$$AB = \frac{1}{2}(SR + PQ)$$



- Area =  $\frac{1}{2} \times (\text{Sum of parallel sides}) \times (\text{height}) = \frac{1}{2} (AB + CD) \times h$
- If non parallel sides are equal then diagonal will also be equal to each other
- If a trapezium is inscribed in a circle then trapezium will be isosceles trapezium (non parallel sides will be equal)
- If non-parallel sides are equal then trapezium is isosceles
- Diagonals intersect each other proportionally in the ratio of the length of parallel side and height

$$\frac{AO}{OC} = \frac{BO}{DO} = \frac{AB}{CD} = \frac{FO}{EO}$$

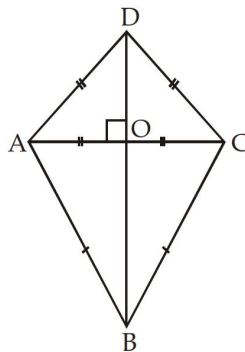


- $AC^2 + BD^2 = BC^2 + AD^2 + 2AB \cdot CD$

**Kite :** Kite is a quadrilateral whose two pair of adjacent sides are equal and the diagonals bisect each other at  $90^\circ$

$$AB = BC, \quad AD = CD$$

$$DB = d_1, \quad AC = d_2$$

**Properties :**

- shorter diagonal is bisected by longer diagonal
- Diagonals intersect at  $90^\circ$
- Pair of adjacent sides are equal

- Area =  $\frac{1}{2} \times \text{product of diagonals}$

## Foundation

### Questions

1. In any parallelogram ABCD, if diagonals cut each other at O and  $OA = 3$  cm,  $OB = 4$  cm, then find the length of line segment BD.
  - (a) 6 cm
  - (b) 8 cm
  - (c) 12 cm
  - (d) 16 cm
2. In a parallelogram ABCD, angle bisector of A and B cut each other at P.  $m \angle APB$  will be ?
  - (a)  $60^\circ$
  - (b)  $180^\circ$
  - (c)  $100^\circ$
  - (d)  $90^\circ$
3. In a trapezium ABCD, intersecting point of AC and BD is at O,  $AB \parallel CD$  and  $AB = 2 \times CD$ . If area of  $\triangle AOB$  is  $84 \text{ cm}^2$ , then find the area of  $\triangle COD$  ?
  - (a)  $21 \text{ cm}^2$
  - (b)  $19 \text{ cm}^2$
  - (c)  $15 \text{ cm}^2$
  - (d)  $13 \text{ cm}^2$
4. ABCD is a trapezium in which  $\overline{AD}$  is parallel to  $\overline{BC}$ . Its diagonals  $\overline{AC}$  and  $\overline{BD}$  cut at point O. Therefore, if  $\overline{AO} = 3$ ,  $\overline{CO} = x - 3$ ,  $\overline{BO} = 3x - 19$ ,  $\overline{DO} = x - 5$ , then find the value of x ?
  - (a) 6, 2
  - (b) 3, 4
  - (c) 8, 9
  - (d) 4, 3
5. Length of diagonal BD of a parallelogram ABCD is 18 cm. If P and Q are respectively centroid of  $\triangle ABC$  and  $\triangle ADC$ , then find the length of PQ ?
  - (a) 3 cm
  - (b) 6 cm
  - (c) 8 cm
  - (d) 9 cm
6. ABCD is a rhombus, whose side  $AB = 4$  cm and  $\angle ABC = 120^\circ$ , therefore, find the length of diagonal BD ?
  - (a) 4 cm
  - (b) 6 cm
  - (c) 8 cm
  - (d) 2 cm
7. If all sides of a parallelogram are equal and one angle is right angle, then figure is called?
  - (a) Square
  - (b) Rectangle
  - (c) Trapezium
  - (d) Kite
8. In quadrilateral ABCD, If  $AB = BC = CD = DA$  and  $AC \neq BD$ , then that quadrilateral is called ?
  - (a) trapezium
  - (b) square
  - (c) rhombus
  - (d) cyclic quadrilateral
9. ABCD is a rectangle and APB is right angled triangle and  $\angle PAB = 60^\circ$  find  $m \angle PDC$  ?
  - (a)  $15^\circ$
  - (b)  $20^\circ$
  - (c)  $30^\circ$
  - (d)  $10^\circ$
10. In the figure,  $AB \parallel DC$ ,  $m \angle ADC$ :
 
  - (a)  $70^\circ$
  - (b)  $60^\circ$
  - (c)  $45^\circ$
  - (d)  $75^\circ$
11. PQRS is a parallelogram. A is point of side PQ. Joining SA and RA such that it bisects  $\angle PSR$  and  $\angle QRS$ , then SR is equal to -
  - (a) PS
  - (b) 2QR
  - (c) QR
  - (d) 4PQ
12. If ABCD is parallelogram whose diagonals cut each other at O and  $\triangle BCD$  is equilateral  $\triangle$  whose each side is 6 cm, then find the length of diagonal AC -
  - (a)  $3\sqrt{3}$  cm
  - (b)  $6\sqrt{3}$  cm
  - (c)  $3\sqrt{6}$  cm
  - (d) 12 cm
13. Area of rhombus is  $96 \text{ cm}^2$ . If diagonal is 12 cm, then length of side of rhombus is -
  - (a) 10 cm
  - (b) 13 cm
  - (c) 24 cm
  - (d)  $2\sqrt{30}$  cm
14. ABCD is a cyclic quadrilateral. AC is diagonal. Which relation is true in the following -
  - (a)  $\angle A + \angle B = 180^\circ$
  - (b)  $\angle A + \angle B = \angle C + \angle D$
  - (c)  $\angle A + \angle C = \angle B + \angle D$
  - (d)  $\angle A + \angle C = 90^\circ$
15. ABCD is a parallelogram and Q and R are circumcentre of  $\triangle ABC$  and  $\triangle ADC$ , then AQCR will be
  - (a) Rectangle
  - (b) Rhombus
  - (c) Trapezium
  - (d) Square
16. If ABCD is trapezium in which  $AB \parallel DC$ , AC and BD cut each other at E, then -
  - (a)  $DE \cdot EA = EC \cdot BC$
  - (b)  $DE \cdot EA = EC \cdot AB$
  - (c)  $DE \cdot EA = EC \cdot DC$
  - (d)  $DE \cdot EA = EB \cdot EC$
17. The length of AB and CD are 10 cm and 14 cm of trapezium ABCD, Its diagonals bisect each other at O, then  $AO : OC$  will be -
  - (a) 5 : 7
  - (b) 12 : 7
  - (c) 7 : 5
  - (d) 7 : 12

18. In quadrilateral ABCD, producing side BA and DC at E and F and  $\angle ABC = x^\circ$ ,  $\angle ADC = y^\circ$ ,  $\angle BCF = a^\circ$ ,  $\angle DAE = b^\circ$  then  $x + y = ?$
- (a)  $a + b$  (b)  $2a + b$   
(c)  $2b + a$  (d)  $a + 3b$
19. ABCD is a square. A line segment CX cuts AB at X and cuts diagonal BD at O. If  $\angle COD = 70^\circ$  and  $\angle OXA = ?$
- (a)  $x^\circ = 125^\circ$  (b)  $x^\circ = 115^\circ$   
(c)  $x^\circ = 135^\circ$  (d)  $x^\circ = 145^\circ$
20. If area of parallelogram is A whose sides are a and b and area of rectangle is B whose sides are a and b then -
- (a)  $A > B$  (b)  $A = B$   
(c)  $A < B$  (d)  $A \geq B$

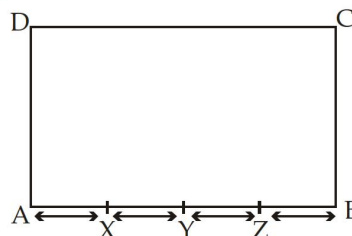
### Moderate

1. ABCED is a quadrilateral. A line from D parallel to diagonal AC meets BC extended at P. Find  $\text{ar}(\triangle ABP)$ ?
- (a)  $\text{ar}(\triangle ACPD)$  (b)  $\text{ar}(\triangle ABCD)$   
(c)  $\text{ar}(\triangle PDA)$  (d)  $\text{ar}(\triangle ABCE)$
2. E, F, G and H are mid-point of any parallelogram ABCD, then EFGH will be also parallelogram and its area will be how many times than area of ABCD?
- (a) 1 (b) 2  
(c)  $\frac{1}{2}$  (d)  $\frac{3}{2}$
3. ABCD is a quadrilateral in which AB is greatest side and CD is smallest side, then relation between  $\angle C$  and  $\angle A$ ?
- (a)  $\angle C = \angle A$  (b)  $\angle C < \angle A$   
(c)  $\angle C > \angle A$  (d)  $\angle C \geq \angle A$
4. ABCD is a square. A line segment DX cuts BC at X and cuts diagonal AC at O such that  $\angle COD = 105^\circ$  and  $\angle OXC = x^\circ$ . Find the value of  $x^\circ$
- (a)  $50^\circ$  (b)  $80^\circ$   
(c)  $60^\circ$  (d)  $70^\circ$
5. ABCD is a square. M is midpoint of AB and N is midpoint of BC. Join DM and AN that form new sides which meet at O. Therefore, which is true in the following?
- (a)  $AB^2 + CD^2 = AD^2$  (b)  $AN = MD$   
(c)  $\angle NAB = \angle ADM$  (d)  $\angle BAN = \angle AMD$
6. ABCD is rhombus. A straight line passing through C meet AD which is produced at P and meet AB produced at Q. Therefore, If  $DP = \frac{1}{2} AB$ , then find the ratio between BQ and AB?
- (a) 2 : 1 (b) 1 : 1  
(c) 1 : 3 (d) 3 : 1
7. Suppose PQRS is a square. M, N and O are points on PQ, QR and RS such that  $PM = QN = RO$ . If  $\triangle MNO$  is right angled  $\triangle$ , then  $\angle MON =$
- (a)  $30^\circ$  (b)  $45^\circ$   
(c)  $60^\circ$  (d)  $90^\circ$
8. MNOP is a trapezium in which  $MN \parallel OP$ . If diagonals cut each other at A, then which is true in the following?
- (a)  $\frac{AM}{AO} = \frac{AN}{AP}$  (b)  $\frac{MP}{NO} = \frac{MN}{OP}$   
(c)  $\frac{AN}{AP} = \frac{NO}{OP}$  (d)  $\frac{MP}{AO} = \frac{MA}{OP}$
9. In trapezium ABCD,  $AB \parallel DC$  and L is mid-point of BC. Draw  $PQ \parallel AD$  from L which meets at P of AB and producing DC at Q. Then  $\text{ar}(\text{trapezium ABCD}) = ?$
- (a)  $\text{ar}(\triangle APQD)$   
(b)  $\text{ar}(\text{quadrilateral APCD})$   
(c)  $\text{ar}(\triangle PLB)$   
(d) None of these
10. ABCD is trapezium In which  $AB \parallel DC$  and  $AB = a$  cm and  $DC = b$  cm. If E and F are mid-points of AD and BC. Then  $\text{ar}(\triangle ABE) : \text{ar}(\triangle FCD) = ?$
- (a)  $a : b$  (b)  $(a + 3b) : (3a + b)$   
(c)  $(3a + b) : (a + 3b)$  (d)  $(2a + b) : (3a + b)$
11. In trapezium ABCD, If  $AB \parallel CD$ , then  $AC^2 + BD^2 = ?$
- (a)  $BC^2 + AD^2 + 2BC \cdot AD$   
(b)  $AB^2 + CD^2 + 2AB \cdot CD$   
(c)  $AB^2 + CD^2 + 2AD \cdot BC$   
(d)  $BC^2 + AD^2 + 2AB \cdot CD$
12. PQRS is a trapezium and  $PQ > RS$  and L, M and N are midpoints of respectively PR, QS and QR then find the length of LM?
- (a)  $\frac{PQ}{2}$  (b)  $\frac{RS}{2}$   
(c)  $\frac{PQ + RS}{2}$  (d)  $\frac{PQ - RS}{2}$
13. In parallelogram ABCD, diagonal AC cuts BE at F. Line BE cuts AD at E, then -
- (a)  $EF \cdot FB = AE \cdot FC$  (b)  $BF \cdot FA = EF \cdot FC$   
(c)  $AE \cdot FC = BC \cdot AF$  (d)  $AE \cdot AB = BC \cdot FB$
14. ABCD is parallelogram. E is mid-point of BC. Producing DE and AD and it meets at F, then which relation is true about AF and AB:
- (a)  $AF^2 = 2AB^2$  (b)  $AF = \frac{3}{2} AB$   
(c)  $AF = 2AB$  (d)  $AF = 3AB$



15. ABCD is a trapezium.  $AB \parallel CD$  and  $AB = 2CD$ . If its diagonals meet at P, then  $\triangle APB : \triangle CPD$  is equal to -  
 (a) 2 : 1 (b) 1 : 2  
 (c) 1 : 4 (d) 4 : 1
16. ABCD is parallelogram and E is mid-point of BC and producing DE and AB which meet at F, then -  
 (a)  $AF = \frac{3}{2} AB$  (b)  $AF^2 = 4AB^2$   
 (c)  $AF = 3AB$  (d)  $AF^2 = 2AB^2$
17. In a trapezium ABCD, If midpoint of diagonals AC and BD are E and F, then EF = ?  
 (a)  $\frac{1}{2} AB$  (b)  $\frac{1}{2} CD$   
 (c)  $\frac{1}{2}(AB + CD)$  (d)  $\frac{1}{2}(AB - CD)$
18. ABCD is a cyclic quadrilateral in which AD and BC are parallel. If  $\angle ABC = 72^\circ$ , the  $m\angle BCD$  ?  
 (a)  $162^\circ$  (b)  $18^\circ$   
 (c)  $108^\circ$  (d)  $72^\circ$
19. ABCD is rhombus in which  $AB = 4$  cm and  $\angle ABC = 120^\circ$ , therefore find the length of diagonal BD ?  
 (a) 1 cm (b) 2 cm  
 (c) 3 cm (d) 4 cm
20. ABCD is a rhombus. In which a straight line passing through C meet P when AD is produced to P and producing AB at Q. Therefore. If  $DP = \frac{1}{2} AB$  then find the ratio of length of BQ and AB ?  
 (a) 2 : 1 (b) 1 : 2  
 (c) 1 : 1 (d) 3 : 1
21. In a parallelogram ABCD, side AB is producing at E such that  $BE = AB$ . Then side DE cuts BC at Q, then Q cuts BC in which ratio?  
 (a) 1 : 2 (b) 1 : 1  
 (c) 2 : 3 (d) 2 : 1
22. ABCD is a quadrilateral whose sides are unequal, if diagonal AC and BD intersect at right angle, then -  
 (a)  $AB^2 + BC^2 = CD^2 + DA^2$   
 (b)  $AB^2 + CD^2 = BC^2 + DA^2$   
 (c)  $AB^2 + AD^2 = BC^2 + CD^2$   
 (d)  $AB^2 + BC^2 = 2(CD^2 + DA^2)$
23. ABCD is a square. M is mid-point of AB and N is mid-point of BC. Join DM and AN form new sides which meet at O. If  $AB = x$  then, value of AN?  
 (a)  $AN = \frac{\sqrt{5}}{4} x$  cm (b)  $AN = \frac{\sqrt{5}}{2} x$  cm  
 (c)  $AN = \frac{\sqrt{3}}{2} x$  cm (d)  $AN = \frac{\sqrt{3}}{4} x$  cm
24. In a trapezium ABCD, AB and DC are parallel and  $AB = 2DC$  and diagonals meet at O, then find the relation between area of  $\triangle OAB$  and  $\triangle OCD$  ?  
 (a) 2 : 1 (b) 4 : 1  
 (c) 1 : 2 (d) 1 : 4

25. In a rectangle ABCD, divide line segment AB through X, Y, Z in four equal parts. Therefore,  $\frac{\text{area}(\triangle XYC)}{\text{area}(\triangle ABCD)}$  will be



- (a)  $\frac{1}{7}$  (b)  $\frac{1}{6}$   
 (c)  $\frac{1}{9}$  (d)  $\frac{1}{8}$
26. In a square PQRS, an equilateral triangle  $\triangle TQR$  is formed, then  $m\angle PTS$  -  
 (a)  $75^\circ$  (b)  $90^\circ$   
 (c)  $120^\circ$  (d)  $150^\circ$
27. ABCD is rhombus. If AB is produced at F and BA is produced at E such that  $AB = AE = BF$ , then result will be -  
 (a)  $ED \parallel CF$  (b)  $ED > CF$   
 (c)  $ED \perp CF$  (d)  $ED^2 + CF^2 = EF^2$
28. If ABCD is a rectangle and P, Q, R, S are respectively mid-points of  $\overline{AB}$ ,  $\overline{BC}$ ,  $\overline{CD}$  and  $\overline{DA}$ , then area of quadrilateral PQRS will be equal to -  
 (a) area of ABCD (b)  $\frac{1}{3}$  area of (ABCD)  
 (c)  $\frac{3}{4}$  area of (ABCD) (d)  $\frac{1}{2}$  area of (ABCD)
29. The parallel sides of a trapezium are p and q respectively. The line joining the mid point of its non-parallel sides will be  
 (a)  $\sqrt{pq}$  (b)  $\frac{2pq}{p+q}$   
 (c)  $\frac{(p+q)}{2}$  (d)  $\frac{1}{2}(p-q)$
30. In a trapezium, the two non-parallel sides are equal in length, each being of 5 cm. The parallel sides are at a distance of 3 cm. If the smaller side of the parallel side is of length 2 cm., then the length of the diagonal of the trapezium is:  
 (a)  $3\sqrt{5}$  cm (b)  $6\sqrt{5}$  cm  
 (c)  $4\sqrt{5}$  cm (d) 7 cm

### Previous Year (Memory Based)

1. ABCD is parallelogram and its diagonals AC and BD cut each other at O. If  $\angle DAC = 32^\circ$  and  $\angle AOB = 70^\circ$ , then  $m\angle DBC$  ?  
 (a)  $30^\circ$  (b)  $102^\circ$   
 (c)  $38^\circ$  (d)  $48^\circ$
2. By joining Mid-points of consecutive sides of any quadrilateral form?  
 (a) rhombus (b) square  
 (c) rectangle (d) None of these
3. PQRS is a rhombus and A is mid-point of PQ.  $RA \perp PQ$ , find  $\angle P$  -  
 (a)  $120^\circ$  (b)  $105^\circ$   
 (c)  $90^\circ$  (d)  $60^\circ$
4. ABCD is rhombus in which  $\angle C = 60^\circ$ , then  $AC : BD = ?$   
 (a)  $\sqrt{3} : 1$  (b)  $\sqrt{3} : \sqrt{2}$   
 (c)  $3 : 1$  (d)  $3 : 2$
5. ABCD is a quadrilateral in which diagonal AC divides it into two equal area, then ABCD will be -  
 (a) Rectangle (b) Parallelogram  
 (c) Rhombus (d) All of these
6. For rectangle ABCD, which is not true ?  
 (a)  $AC = BD$  (b)  $AB + BC > BD$   
 (c)  $CD + DA < BD$  (d)  $AB \neq AC$
7. If sides of quadrilateral ABCD touch a circle, then  $AB + CD$  is equal to -  
 (a)  $2AD$  (b)  $2BC$   
 (c)  $AD + BC$  (d)  $AD - BC$
8. In a quadrilateral ABCD, M is mid-point of BD. MB is bisector of  $\angle B$ .  $m\angle AMB$  ?  
 (a)  $45^\circ$  (b)  $60^\circ$   
 (c)  $90^\circ$  (d)  $120^\circ$
9. ABCD is a quadrilateral in which AC and BD are diagonals and diagonals intersect at O, then -  
 $AB + BC + CD + DA$   
 (a)  $> 2(AC + BD)$  (b)  $> AC + BD$   
 (c)  $< AC + BD$  (d)  $< 2(AC + BD)$
10. ABCD is quadrilateral in which AB is greatest side and CD is smallest side then -  
 (a)  $\angle C > \angle A, \angle D > \angle B$  (b)  $\angle C > \angle B, \angle A > \angle D$   
 (c)  $\angle C > \angle D, \angle A > \angle B$  (d)  $\angle C + \angle D > \angle B + \angle A$
11. ABCD is a trapezium in which  $AB \parallel DC$  and E is mid-point of AD. If F is any point on BC in which  $EF \parallel AB$ , then  $EF = ?$   
 (a)  $AB + DE$  (b)  $AB + CD$   
 (c)  $2AB + CD$  (d)  $\frac{1}{2}(AB + CD)$
12. Parallel sides of trapezium are a and b. line joining the mid points of the non-parallel sides will be -  
 (a)  $\frac{1}{2}(a - b)$  (b)  $\frac{1}{2}(a + b)$   
 (c)  $\frac{2ab}{a + b}$  (d)  $\sqrt{ab}$
13. Length of diagonal of parallelogram ABCD is 36 cm. If M and N are centroid of  $\triangle ABC$  and  $\triangle ADC$ , then length of line segment MN is ?  
 (a) 6 cm (b) 8 cm  
 (c) 9 cm (d) 12 cm
14. Length of diagonal BD of a parallelogram ABCD is 18 cm. If P and Q are centroid of  $\triangle ABC$  and  $\triangle ADC$ , then length of PQ will be -  
 (a) 4 cm (b) 6 cm  
 (c) 9 cm (d) 12 cm
15. Perimeter of a rhombus is 40 cm and one angle is  $60^\circ$ . Find the area of rhombus ?  
 (a)  $100\sqrt{3} \text{ cm}^2$  (b)  $50\sqrt{3} \text{ cm}^2$   
 (c)  $160\sqrt{3} \text{ cm}^2$  (d)  $100 \text{ cm}^2$
16. ABCD is a rhombus in which ratio of  $\angle A$  and  $\angle B$  is 4:5, then  $m\angle C$ ?  
 (a)  $50^\circ$  (b)  $45^\circ$   
 (c)  $80^\circ$  (d)  $95^\circ$
17. In a parallelogram ABCD, one side  $AB = 24$  cm and second side  $AD = 16$  cm. Distance between AB and DC is 10 cm. Therefore, distance between AD and BC will be ?  
 (a) 16 cm (b) 18 cm  
 (c) 15 cm (d) 26 cm
18. ABCD is a trapezium in which  $AB = CD$ ,  $AD \parallel BC$ ,  $AD = 5$  cm and  $BC = 9$  cm. therefore, If area of ABCD is  $35 \text{ cm}^2$ , then find the length of CD?  
 (a)  $\sqrt{29}$  cm (b) 5 cm  
 (c) 6 cm (d)  $\sqrt{21}$  cm
19. ABCD is a quadrilateral in which diagonal  $BD = 64$  cm,  $AL \perp BD$  and  $CM \perp BD$ ,  $AL = 13.2$  cm and  $CM = 16.8$  cm, find the area of ABCD (in  $\text{cm}^2$ ) ?  
 (a) 422.4 (b) 690  
 (c) 537.6 (d) 960
20. If side of quadrilateral ABCD touches a circle and  $AB = 6$  cm,  $CD = 5$  cm,  $BC = 7$  cm, then AD in cm is -  
 (a) 8 (b) 9  
 (c) 4 (d) 6

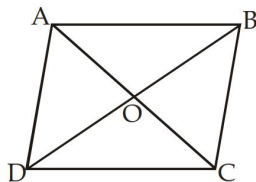


21. ABCD is a trapezium with parallel sides  $AB = 2$  cm and  $DC = 3$  cm. E and F are the mid-points of the non-parallel sides. The ratio of the area of ABFE to the area of EFCD is :
- (a)  $\frac{9}{11}$  (b)  $\frac{3}{4}$   
 (c)  $\frac{7}{11}$  (d)  $\frac{5}{7}$
22. Three angles of a quadrilateral are  $60^\circ$ ,  $120^\circ$  and  $90^\circ$ , then find the fourth angle in radian ?
- (a)  $\frac{\pi}{4}$  (b)  $\frac{3\pi}{4}$   
 (c)  $\frac{\pi}{3}$  (d)  $\frac{\pi}{2}$
23. Area of a parallelogram, a rhombus and a triangle on the same base and between the same parallel lines are P, R and T respectively. Then which one is true ?
- (a)  $R < P < T$  (b)  $P > R > T$   
 (c)  $R = P = T$  (d)  $R = P = 2T$
24. ABCD is a parallelogram in which  $AB = 10$  cm,  $AD = 6$  cm. Bisector of  $\angle A$  meet at E through DC and meet F through BC. Therefore, length of CF will be ?
- (a) 4 cm (b) 2 cm  
 (c) 6 cm (d) 8 cm
25. ABCD is a parallelogram. BC is produced to Q such that  $BC = CQ$ , then -
- (a) area  $(\triangle BCP) = \text{Area}(\triangle DPQ)$   
 (b) area  $(\triangle BCP) > \text{area}(\triangle DPQ)$   
 (c) area  $(\triangle BCP) < \text{area}(\triangle DPQ)$   
 (d) area  $(\triangle BCP) + \text{area}(\triangle DPQ) = \text{area}(ABCD)$
26. All sides of a quadrilateral ABCD touch a circle. In which if  $AB = 6$  cm,  $BC = 7.5$  cm,  $CD = 3$  cm, then find the length of DA ?
- (a) 3.5 cm (b) 4.5 cm  
 (c) 2.5 cm (d) 1.5 cm
27. ABCD is a rectangle in which ratio of AB and BC is 3:2. If P is mid-point of AB, then value of  $\sin \angle CPB$  ?
- (a)  $\frac{3}{5}$  (b)  $\frac{1}{5}$   
 (c)  $\frac{3}{4}$  (d)  $\frac{4}{5}$
28. ABCD is a quadrilateral that touches a circle in which  $AB = 6$  cm,  $CD = 5$  cm and  $AD = 7$  cm. Therefore, find the length of BC ?
- (a) 6 cm (b) 4 cm  
 (c) 5 cm (d) 3 cm
29. In a trapezium ABCD,  $BC \parallel AD$ ,  $\angle BAE = 30^\circ$ ,  $\angle CDF = 45^\circ$ ,  $BC = 6$  cm and  $AB = 12$  cm. Find the area of ABCD.
- (a)  $18(2 + \sqrt{3}) \text{ cm}^2$  (b)  $36\sqrt{3} \text{ cm}^2$   
 (c)  $12(3 + 2\sqrt{3}) \text{ cm}^2$  (d) None of these
30. What is the ratio of area of square of side 'a' and square formed on its diagonal ?
- (a) 1 : 2 (b) 1 : 4  
 (c) 2 : 1 (d) 4 : 1

## Foundation

### Solutions

1. (b); Diagonals of parallelogram bisect each other.

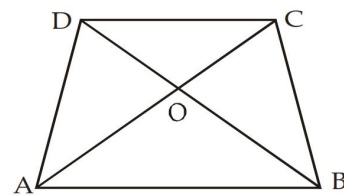


$$\therefore OC = OA = 3 \text{ cm}, \quad OD = OB = 4 \text{ cm}$$

$$AC = 2 \times OA = 6 \text{ cm} \quad BD = 2 \times OB = 8 \text{ cm}$$

2. (d);  $\angle APB = 90^\circ$  (because in parallelogram sum of two consecutive angle =  $180^\circ$  and angle bisector of  $(A+B) = 90^\circ$   
 so, third vertex will be  $90^\circ$

3. (a); In  $\triangle AOB$  and  $\triangle COD$ ,



$$\angle OAB = \angle OCD$$

(alternate interior angle)

$$\angle OBA = \angle ODC$$

(alternate interior angle)

$$\therefore \triangle AOB \sim \triangle COD$$

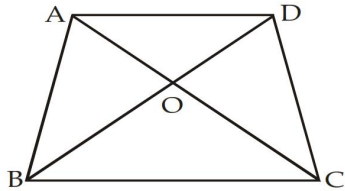
(by AA - similarity)

$$\frac{\text{area}(\triangle AOB)}{\text{area}(\triangle COD)} = \frac{AB^2}{CD^2} = \frac{(2CD)^2}{CD^2} = \frac{4CD^2}{CD^2} = 4$$

$$\therefore \text{Area}(\triangle COD) = \frac{1}{4} \times \text{area}(\triangle AOB)$$

$$= \frac{1}{4} \times 84 = 21 \text{ cm}^2$$

4. (c); In  $\triangle AOB$  and  $\triangle BOC$ ,



$$\frac{OA}{OC} = \frac{OB}{OD}$$

$$\frac{3}{x-3} = \frac{x-5}{3x-19}$$

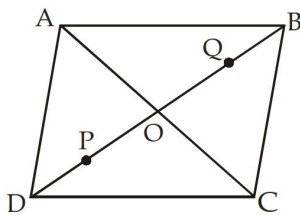
$$\Rightarrow 3(3x-19) = (x-3)(x-5)$$

$$\Rightarrow 9x - 57 = x^2 - 8x + 15$$

$$\Rightarrow x^2 - 17x + 72 = 0$$

$$\therefore x = 8, 9$$

5. (b); OB and OD are medians.



Diagonal divides parallelogram into two congruent triangles

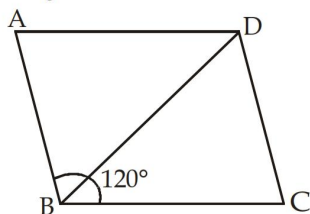
$$\frac{DP}{PO} = \frac{2}{1}, \frac{BQ}{OQ} = \frac{2}{1}$$

and  $PO = OQ$ .

$$\text{so, } PO = \frac{1}{3} \times 9 = 3$$

$$OQ = 3, PO = OQ = 6$$

6. (a); Diagonals of rhombus bisect each angle.



$$\therefore \angle ABD = \frac{120^\circ}{2} = 60^\circ$$

$$\therefore \angle ABD = \angle ADB = 60^\circ$$

$$\therefore AB = AD$$

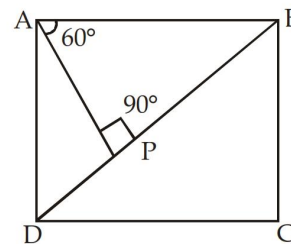
$$\therefore \triangle ABD \text{ will be equilateral triangle}$$

$$\therefore BD = 4 \text{ cm}$$

7. (a);

8. (c);

9. (c);



$$\angle PDC = 90^\circ - \angle ADP$$

$$\angle ADP = 180^\circ - \angle DAP - \angle APD$$

$$= 180^\circ - 30^\circ - 90^\circ$$

$$\angle ADP = 60^\circ$$

$$\angle PDC = 90^\circ - \angle ADP = 90^\circ - 60^\circ$$

$$\angle PDC = 30^\circ$$

10. (b);  $\angle B + \angle C = 180^\circ$

$$\angle C = 180^\circ - 90^\circ = 90^\circ$$

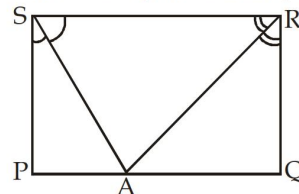
In quadrilateral AECD,

$$\angle A + \angle E + \angle C + \angle D = 360^\circ$$

$$\Rightarrow 90^\circ + (180^\circ - 60^\circ) + 90^\circ + \angle D = 360^\circ$$

$$\therefore \angle D = 60^\circ$$

11. (b);  $\angle ARS = \angle QRA = \angle RAQ$



$$\therefore RQ = AQ$$

And  $\angle PSA = \angle ASR = \angle SAP$

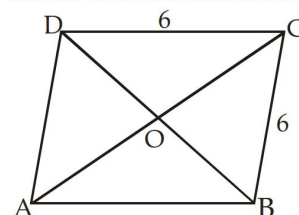
$$\therefore PS = PA$$

$$\therefore PS + QR = PA + AQ$$

$$\therefore 2QR = SR$$

12. (b);  $AB = BC = CD = AD = 6 \text{ cm}$

$\therefore$  ABCD is also a rhombus. Diagonals of rhombus bisect each other at  $90^\circ$ .



$$\therefore OD = \frac{1}{2} BD = 3 \text{ cm}$$

$$\therefore OC = \sqrt{6^2 - 3^2} = \sqrt{27} = 3\sqrt{3} \text{ cm}$$

$$\begin{aligned}\therefore AC &= 2 OC \\ &= 2 \times 3\sqrt{3} = 6\sqrt{3} \text{ cm}\end{aligned}$$

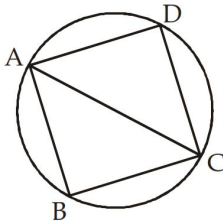
13. (a); Area of rhombus =  $\frac{1}{2}$  × product of diagonals

$$96 = \frac{1}{2} \times 12 \times \text{second diagonal}$$

$$\therefore \text{Second diagonal} = 16 \text{ cm}$$

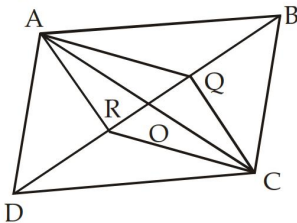
$$\begin{aligned}\text{Length of side} &= \sqrt{\left(\frac{16}{2}\right)^2 + \left(\frac{12}{2}\right)^2} = \sqrt{8^2 + 6^2} \\ &= 10 \text{ cm}\end{aligned}$$

14. (c); ABCD is a cyclic quadrilateral.



$$\therefore \angle A + \angle C = \angle B + \angle D$$

15. (b); Q is circumcentre of



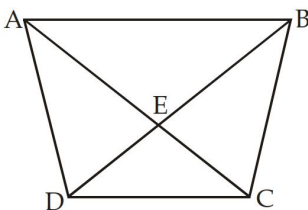
$$\therefore QO \text{ is perpendicular of } AC.$$

So, RO is also perpendicular of AC and  $\triangle ABC$  and  $\triangle ADC$  is congruent.

$$\therefore AC \text{ and } QR \text{ are perpendicular to each other}$$

$$\therefore AQCR \text{ is a rhombus.}$$

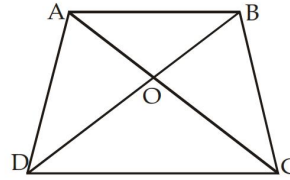
16. (d);  $\triangle DEC \sim \triangle BEA$



$$\therefore \frac{DE}{EB} = \frac{EC}{EA}$$

$$\therefore DE \cdot EA = EB \cdot EC$$

17. (a);  $\triangle AOB \sim \triangle COD$



$$\therefore \frac{AO}{CO} = \frac{AB}{CD}$$

$$= \frac{10}{14} = \frac{5}{7} \Rightarrow 5:7$$

18. (a);  $\angle A + b^\circ = 180^\circ$  (linear pair) ... (i)

$\angle C + a^\circ = 180^\circ$  (linear pair) ... (ii)

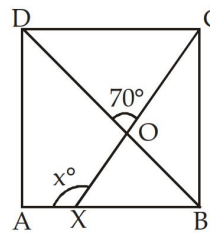
From (i) and (ii)

$$\angle A + \angle B + \angle C + \angle D = 360^\circ$$

$$180^\circ - b^\circ + x^\circ + 180^\circ - a^\circ + y^\circ = 360^\circ$$

$$x + y = a + b$$

19. (b);



In  $\triangle XOB$ ,

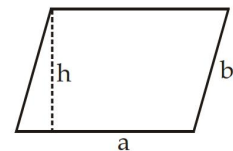
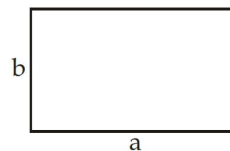
$$\angle XOB + \angle BXO + \angle OXB = 180^\circ$$

$$70^\circ + \angle BXO + 45^\circ = 180^\circ$$

$$\angle BXO = 65^\circ$$

$$\angle OXA = 180^\circ - 65^\circ = 115^\circ$$

20. (c); Let height of parallelogram = h



$$h < b$$

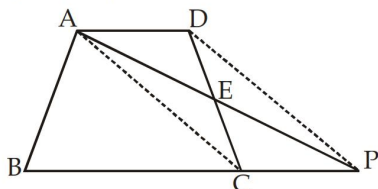
$$A = a \times h, B = a \times b$$

$$\text{so, } A < B$$



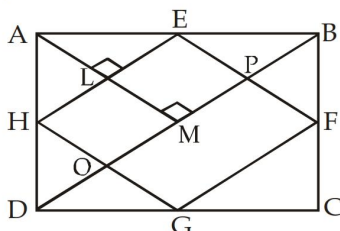
## Moderate

1. (b);  $\therefore DP \parallel AC$



$$\begin{aligned} \therefore \text{ar}(\triangle ACD) &= \text{ar}(\triangle ACP) \\ \text{Adding both side ar}(\triangle ABC), \\ \text{ar}(\triangle ABC) + \text{ar}(\triangle ACD) &= \text{ar}(\triangle ABC) + \text{ar}(\triangle ACP) \\ \therefore \text{ar}(ABCD) &= \text{ar}(\triangle ABP) \end{aligned}$$

2. (C); Join B and D.



Draw  $AM \perp BD$ , which cuts  $HE$  at  $L$ .

In  $\triangle ABD$ ,

Midpoint of  $AB$  is  $E$  and midpoint of  $AD$  is  $H$ ;

$$\therefore HE \parallel DB \text{ and } HE = \frac{1}{2} DB$$

similarly,  $GF \parallel DB$

$$\therefore EFGH \text{ is parallelogram}$$

In  $\triangle ADM$ ,

Midpoint of  $AD$  is  $H$  and  $HL \parallel DM$

$$\therefore AL = LM = \frac{1}{2} AM$$

$$\text{ar}(\triangle ABD) = \frac{1}{2} \times BD \times AM$$

$$\text{ar}(\triangle HEPO) = EH \times LM = \frac{1}{2} BD \times \frac{1}{2} AM$$

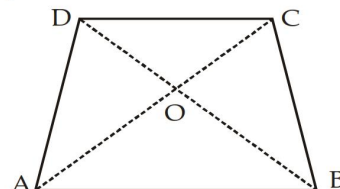
$$= \frac{1}{2} \left( \frac{1}{2} BD \times AM \right) = \frac{1}{2} \text{ar}(\triangle ABD)$$

$$\text{similarly, area}(\triangle OPFG) = \frac{1}{2} \text{ar}(\triangle CBD)$$

$$\text{area of } (EFGH) = \frac{1}{2} \text{ar}(\triangle ABD) + \frac{1}{2} \text{ar}(\triangle CBD)$$

$$= \frac{1}{2} \text{ar}(ABCD)$$

3. (c); Join  $AC$  and  $BD$ .



In  $\triangle ABC$ ,

$AB > BC$

$$\therefore \angle ACB > \angle BAC$$

In  $\triangle ADC$ ,

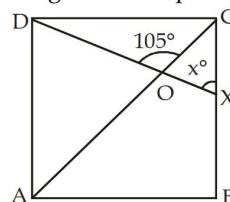
$AD > DC$

$$\therefore \angle ACD > \angle CAD$$

$$\therefore \angle ACB + \angle ACD > \angle BAC + \angle CAD$$

$$\therefore \angle C > \angle A$$

4. (c); Diagonal of a square bisects angle.



$$\therefore \angle OCX = 45^\circ$$

( $\therefore \angle DCB = 90^\circ$  and  $CA$  bisects  $\angle DCB$ )

$\angle COD + \angle COX = 180^\circ$  (linear pair)

$$105^\circ + \angle COX = 180^\circ, \angle COX = 75^\circ$$

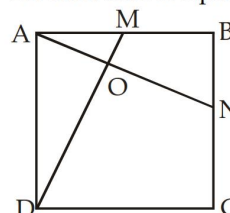
In  $\triangle COX$ ,

$$\angle OCX + \angle COX + \angle OXC = 180^\circ$$

$$45^\circ + 75^\circ + \angle OXC = 180^\circ$$

$$\therefore \angle OXC = 60^\circ$$

5. (b); Let each side of square =  $x$



$$\therefore AB = BC = CD = AD = x$$

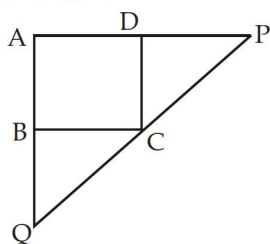
$$BN = AM = \frac{x}{2}$$

$$AN = \sqrt{x^2 + \left(\frac{x}{2}\right)^2} = \sqrt{x^2 + \frac{x^2}{4}} = \sqrt{\frac{5x^2}{4}}$$

$$DM = \sqrt{x^2 + \left(\frac{x}{2}\right)^2} = \sqrt{\frac{5x^2}{4}} \text{ so, } AN = DM$$

6. (a); Let each side of rhombus = x

Let BQ = a



$$\therefore AB = BC = CD = AD = x$$

$$DP = \frac{1}{2} AB$$

$$\therefore DP = \frac{x}{2}$$

$$\triangle APQ \sim \triangle DPC$$

$$\frac{AP}{DP} = \frac{AQ}{CD}, \quad \frac{x + \frac{x}{2}}{\frac{x}{2}} = \frac{x+a}{x}$$

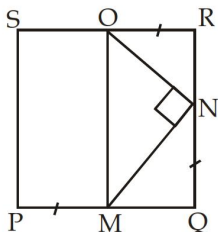
$$\Rightarrow 3x = x + a$$

$$\therefore a = 2x$$

$$\therefore BQ : AB$$

$$= a : x = 2x : x = 2 : 1$$

7. (b);



$$PM = QN$$

$$MQ = NR$$

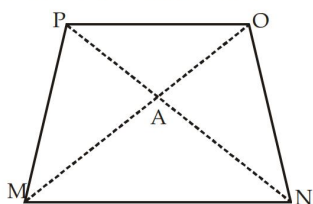
$$MN = NO$$

$$\therefore \angle NOM = \angle NMO$$

$$\therefore \angle NOM + \angle NMO = 90^\circ$$

$$\therefore \angle NOM = 45^\circ$$

8. (a); From  $\triangle AMN$  and  $\triangle APO$



$$\angle OAP = \angle NAM \text{ (Vertically opposite angle)}$$

$$\angle ANM = \angle APO \text{ (Alternate interior angle)}$$

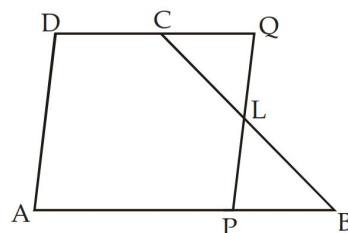
$$\angle AOP = \angle AMN \text{ (Alternate interior angle)}$$

$$\therefore \triangle AMN \sim \triangle AOP$$

$$\Rightarrow \frac{AO}{AM} = \frac{AP}{AN} = \frac{OP}{MN}$$

$$\therefore \frac{AM}{AO} = \frac{AN}{AP} = \frac{MN}{OP}$$

9. (a); Form  $\triangle LPB$  and  $\triangle LQC$ ,



$$BL = CL,$$

$$\angle BPL = \angle CQL$$

$$\angle PBL = \angle QCL$$

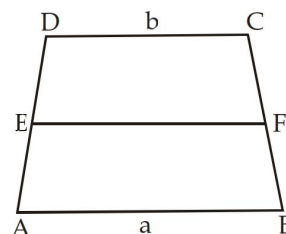
$$\therefore \triangle LPB \cong \triangle LQC$$

$$\therefore \text{ar (trapezium ABCD)}$$

$$= \text{ar (APQD)} + \text{ar (PBL)} - \text{ar (QCL)}$$

$$= (\text{||gm APQD})$$

10. (c);  $EF = \frac{1}{2} (a + b)$



Let distance between AB and EF = d

$$\therefore \text{ar (trapezium ABEF)}$$

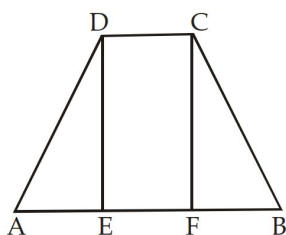
$$= \frac{1}{2} \left( a + \frac{a+b}{2} \right) \times d = \frac{(3a+b)}{4} d$$

$$\text{ar (trapezium EFCD)} = \frac{1}{2} \left( \frac{a+b}{2} + b \right) \times d$$

$$= \frac{(a+3b)}{4} d$$

$$\therefore \text{ratio} = \frac{(3a+b)d}{4} : \frac{(a+3b)d}{4} = (3a+b) : (a+3b)$$

11. (d); In  $\triangle ABC$



$\angle B$  is acute angle.

$$AC^2 = BC^2 + AB^2 - 2AB \cdot FB \quad \dots(i)$$

In  $\triangle ABD$ ,

$\angle A$  is acute angle.

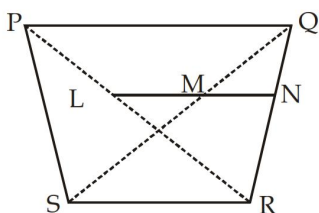
$$\therefore BD^2 = AD^2 + AB^2 - 2AB \cdot AE \quad \dots(ii)$$

From equation (i) + equation (ii),

$$(AC)^2 + (BD)^2 = BC^2 + AD^2 + 2AB(AB - FB - AE) \\ = BC^2 + AD^2 + 2AB \cdot CD.$$

12. (d);  $PQ \parallel RS$

$$\therefore LM \parallel PQ \parallel RS$$



From  $\triangle PQR$ ,

$LN \parallel PQ$

$$\therefore LN = \frac{PQ}{2} \quad \dots(i)$$

From  $\triangle QRS$ ,

$MN \parallel RS$

$$\therefore MN = \frac{SR}{2} \quad \dots(ii)$$

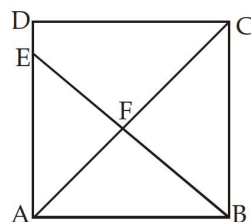
$$\Rightarrow LN - LM = \frac{SR}{2}$$

$$\Rightarrow \frac{QR}{2} - LM = \frac{SR}{2} \text{ [using (i)]}$$

$$\Rightarrow LM = \frac{QP}{2} - \frac{SR}{2}$$

$$\therefore LM = \frac{QP - SR}{2}$$

13. (b); In  $\triangle$  and  $\triangle CFB$ ,



$$\angle EFA = \angle CFB$$

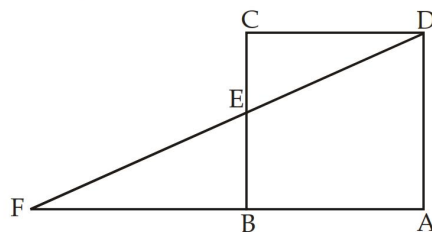
$$\angle EAF = \angle BCF$$

$$\angle AEF = \angle FCB$$

$$\therefore \frac{BF}{FE} = \frac{FC}{FA}$$

$$\therefore BF \cdot FA = FE \cdot FC$$

14. (c);  $\triangle BFE \cong \triangle CDE$



$$\left[ \begin{array}{l} \therefore \angle BFE = \angle CDE, \\ \angle BEF = \angle CED, CE = BE \end{array} \right]$$

$$\therefore FB = CD$$

$$\text{But } CD = BA$$

$$\therefore FB = BA$$

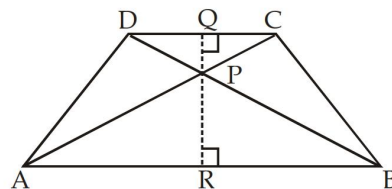
$$\therefore FB + BA = BA + BA$$

$$\Rightarrow FB + BA = 2BA$$

$$\therefore AF = 2AB$$

15. (d);  $AB \parallel CD$  and

$$AB = 2CD$$



$$\triangle APB \sim \triangle CPD$$

$$\text{so, } \frac{PA}{PC} = \frac{PB}{PD} = \frac{AB}{DC} = \frac{PR}{QP}$$

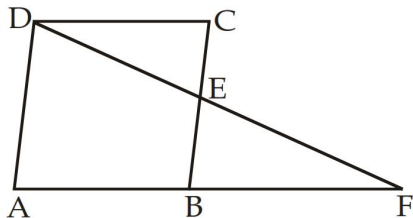
$$\frac{\text{ar}(\triangle APB)}{\text{ar}(\triangle CPD)} = \frac{\frac{1}{2} \times AB \times PR}{\frac{1}{2} \times CD \times PQ}$$



$$= \frac{AB \times PR}{CD \times PQ} = \frac{AB}{CD} \times \frac{AB}{CD} \text{ [using (i)]}$$

$$= \left( \frac{AB}{CD} \right)^2 = \left( \frac{2}{1} \right)^2 = 4 : 1$$

16. (b);  $\triangle EDC \cong \triangle EFB$



$$\angle DEC = \angle BEF$$

$$\angle DCE = \angle EBF \text{ (Alternate angle)}$$

$$\triangle CDE \approx \triangle BFE$$

$$\frac{CD}{BF} = \frac{DE}{FE} = \frac{CE}{BE}, \quad \frac{CE}{BE} = \frac{CD}{BF}$$

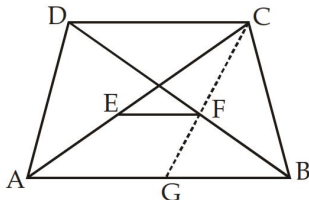
$$CD = BF \text{ (CE = BE)}$$

$$\text{so, } AF = AB + BF$$

$$= AB + CD$$

$$AF = 2AB \Rightarrow AF^2 = 4AB^2$$

17. (d); Join CF and Produce at G on AB.



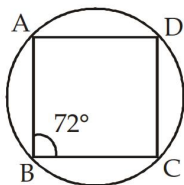
$$\text{Then, } \triangle CDF \cong \triangle GBF$$

$$CD = GB \text{ and } CF = GF$$

E and F are mid points of CA and CG.

$$\therefore EF = \frac{1}{2} AG = \frac{1}{2} (AB + CD)$$

18. (d);  $\angle CDA = 180^\circ - \angle ABC$



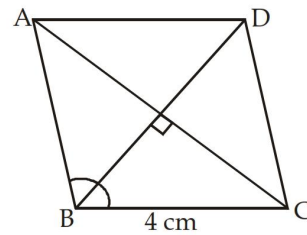
$$= 180^\circ - 72^\circ = 108^\circ$$

$$AD \parallel BC$$

$$\angle BCD + \angle CDA = 180^\circ$$

$$\angle BCD = 180^\circ - 108^\circ = 72^\circ$$

19. (d);  $\angle ABC = 120^\circ$



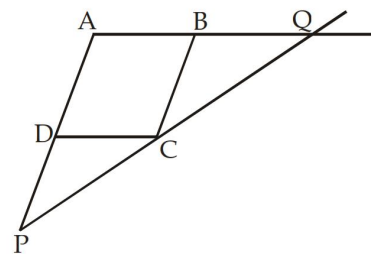
$$\angle DBC = \frac{120}{2} = 60^\circ$$

$$\text{In } \triangle BDC, BC = CD$$

$$\angle DBC = \angle BDC = 60^\circ$$

so,  $\triangle BDC$  is equilateral triangle,  $BD = 4 \text{ cm}$

20. (a); ABCD is rhombus.



$$AB = BC = CD = AD$$

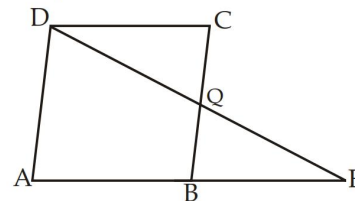
$$DP = \frac{1}{2} AB = \frac{1}{2} BC = \frac{1}{2} CD = \frac{1}{2} DA$$

from similarity

$$\frac{AB + BQ}{BQ} = \frac{AD + DP}{BC}$$

$$\frac{AB}{BQ} = \frac{3}{2} - 1 = \frac{1}{2}, \quad \frac{BQ}{AB} = \frac{2}{1}$$

21. (b);  $AD \parallel BC$



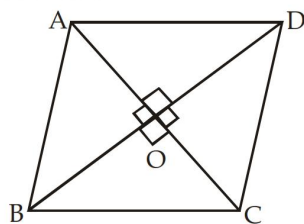
$$AD \parallel BQ$$

$$\triangle DQC \sim \triangle EQB \left[ \begin{array}{l} \angle DQC = \angle BQE \\ \angle DCQ = \angle QBE \\ \angle CDQ = \angle QEB \end{array} \right]$$

$$\frac{DQ}{QE} = \frac{CQ}{BQ} = \frac{CD}{BE} \Rightarrow \frac{CD}{BE} = \frac{CQ}{BQ}$$

$$CQ : BQ = 1 : 1$$

22. (b); In  $\triangle OBC$



$$OB^2 + OC^2 = BC^2$$

In  $\triangle OCD$ ,

$$OC^2 + OD^2 = CD^2$$

In  $\triangle OAD$ ,

$$OD^2 + OA^2 = AD^2$$

In  $\triangle OAB$ ,

$$OA^2 + OB^2 = AB^2$$

(i) + (ii) + (iii) + (iv)

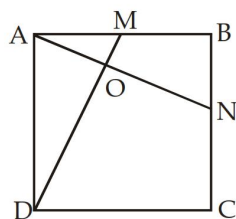
$$2(OB^2 + OA^2 + OD^2 + OC^2) = AB^2 + BC^2 + CD^2 + DA^2$$

$$\Rightarrow 2(AB^2 + CD^2)$$

$$= AB^2 + BC^2 + CD^2 + DA^2$$

$$\Rightarrow AB^2 + CD^2 = BC^2 + DA^2$$

23. (b); Let  $BA = BC = CD = AD = x$  cm



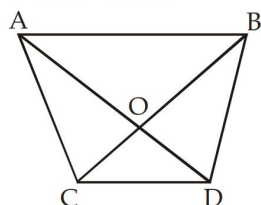
$$BN = NC = \frac{x}{2} \text{ cm}$$

In  $\triangle ABN$ ,

$$AN = \sqrt{x^2 + \left(\frac{x}{2}\right)^2}$$

$$= \frac{\sqrt{5}x}{2} \text{ cm}$$

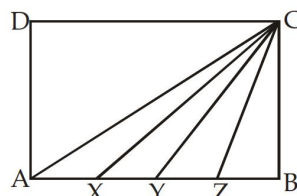
24. (b);  $\triangle OAB \sim \triangle OCD$



$$\frac{\text{area}(\triangle OAB)}{\text{area}(\triangle OCD)} = \frac{AB^2}{DC^2}$$

$$\frac{(2DC)^2}{DC^2} = \frac{4DC^2}{DC^2} = \frac{4}{1}$$

25. (d);  $AX = XY = YZ = BZ$



...(i)

...(ii)

...(iii)

...(iv)

Area of triangle on same base is equal in ratio of base.

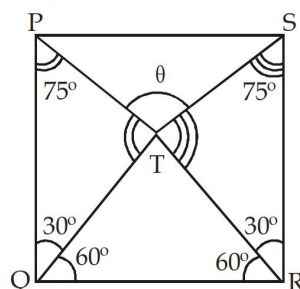
$$\text{area}(\triangle ACX) = \text{area}(\triangle XYC) = \text{area}(\triangle YZC) = \text{area}(\triangle BZC)$$

$$\text{area}(\triangle ABC) = \frac{1}{2} \text{area}(ABCD)$$

$$\text{area}(\triangle XYC) = \frac{1}{8} \text{area}(ABCD)$$

$$\frac{\text{Area}(\triangle XYC)}{\text{Area}(ABCD)} = \frac{1}{8}$$

26. (d); In square PQRS



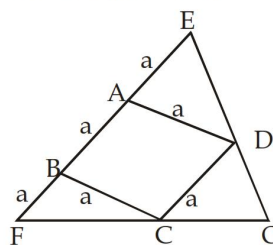
$\triangle PTS$

$$\angle SPT = 90^\circ - 75^\circ = 15^\circ$$

$$\angle PST = 90^\circ - 75^\circ = 15^\circ$$

$$\therefore \angle PTS = 180^\circ - (15^\circ + 15^\circ) = 180^\circ - 30^\circ = 150^\circ$$

27. (c);  $AE = AD$  (In  $\triangle ADE$ )



$$\angle E = \angle D = a$$

$$\angle A = a + a = 2a, \quad \angle C = \angle F = a$$

$$\angle B = d + d = 2d, \quad \angle A + \angle B = 180^\circ$$

$$2a + 2a = 180^\circ, \quad a = 45^\circ$$

In  $\triangle OEF$ ,

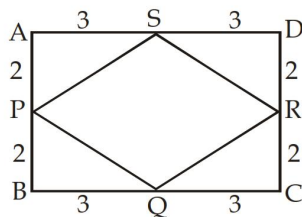
$$\angle O + \angle E + \angle F = 180^\circ$$

$$\angle O + 45 + 45 = 180^\circ$$

$$\angle O = 90^\circ$$

$ED \perp CF$

28. (d); Let sides of ABCD is-



$AD = BC = 6$  unit and

$AB = CD = 4$  unit

$\therefore$  Area ABCD =  $6 \times 4 = 24$  And

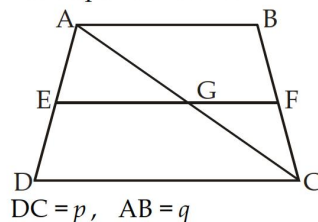
$$\text{Area PQRS} = \text{ABCD} - [\triangle SAP + \triangle PBQ + \triangle QCR + \triangle RDS]$$

$$= 24 - \frac{1}{2} [2 \times 3 + 2 \times 3 + 2 \times 3 + 2 \times 3]$$

$$= 24 - \frac{1}{2} \times 24 = 24 - 12 = 12 \text{ unit}$$

$$\therefore \text{Area PQRS} = \frac{1}{2} \text{ area ABCD}$$

29. (c); In a trapezium ABCD



$$DC = p, \quad AB = q$$

$AB \parallel CD$

In  $\triangle ADC$ ,  $EF \parallel DC$

$$EG = \frac{1}{2} DC \quad \dots(i)$$

Similarly, In  $\triangle ABC$ ,

$AB \parallel GF$

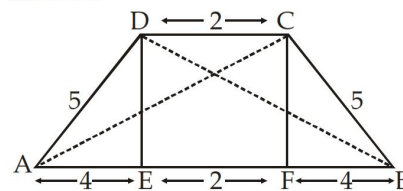
$$GF = \frac{1}{2} AB \quad \dots(ii)$$

(i) + (ii)

$$EG + GF = \frac{1}{2} DC + \frac{1}{2} AB$$

$$EF = \frac{1}{2} (DC + AB), \quad EF = \frac{1}{2} (p + q)$$

30. (a); In  $\triangle ACF$

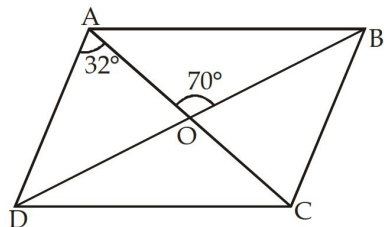


$$AC = \sqrt{6^2 + 3^2} = \sqrt{45} = 3\sqrt{5} \text{ cm}$$

$$\text{In } \triangle BDE, \quad DB = \sqrt{6^2 + 3^2} = 3\sqrt{5} \text{ cm}$$

### Previous Year (Memory Based)

1. (c);  $\angle DAC = 32^\circ$



$$\angle AOB = 70^\circ$$

In  $\triangle AOD$ ,

$$\angle ADO + \angle OAD + \angle AOD = 180^\circ$$

$$\angle ADO + 32^\circ + 180^\circ - \angle AOB = 180^\circ$$

$$\Rightarrow \angle ADO + 32^\circ + 180^\circ - 70^\circ = 180^\circ$$

$$\Rightarrow \angle ADO + 32^\circ + 110^\circ = 180^\circ$$

$$\therefore \angle ADO = 38^\circ$$

ABCD is a parallelogram.

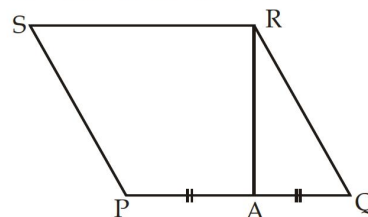
$$\therefore AD \parallel BC$$

$$\therefore \angle ADB = \angle DBC$$

$$\therefore \angle DBC = \angle ADO = 38^\circ$$

2. (d);

3. (a); Let each side of rhombus = x



$$AQ = \frac{x}{2}$$

In  $\Delta ARQ$ ,

$$x^2 = (RA)^2 + \left(\frac{x}{2}\right)^2$$

$$(AR)^2 = x^2 - \frac{x^2}{4} = \frac{3x^2}{4}$$

$$\therefore AR = \frac{\sqrt{3}x}{2}$$

Form  $\Delta ARQ$ ,

$$\sin Q = \frac{AR}{QR} = \frac{\sqrt{3}x}{2x} = \frac{\sqrt{3}}{2}$$

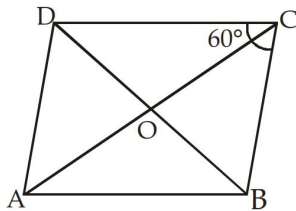
$$\therefore \angle Q = 60^\circ$$

In rhombus PQRS,

$$\angle P + \angle Q = 180^\circ$$

$$\therefore \angle P = 180^\circ - 60^\circ = 120^\circ$$

4. (a); Each side of rhombus is equal.



$$\therefore BC = DC$$

$$\therefore \angle BDC = \angle DBC = x^\circ$$

$$x^\circ + x^\circ + 60^\circ = 180^\circ, \quad x = 60^\circ$$

$$\therefore \angle BDC = \angle DBC = \angle BCD = 60^\circ$$

$$\therefore \Delta BCD \text{ is equilateral triangle.}$$

$$\text{Let } BD = BC = a$$

$$AB^2 = OA^2 + OB^2$$

$$\Rightarrow a^2 - \left(\frac{a}{2}\right)^2 = \frac{3a^2}{4}$$

$$\therefore OA = \frac{\sqrt{3}a}{2}$$

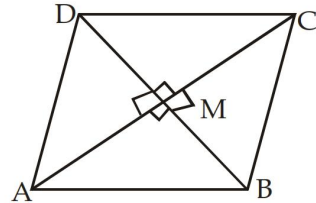
$$\therefore AC = 2 \times \frac{\sqrt{3}}{2}a = \sqrt{3}a$$

$$\therefore AC : BD = \sqrt{3}a : a = \sqrt{3} : 1$$

5. (d);      6. (c);      7. (c);

8. (c); Quadrilateral ABCD is a rhombus or square because MB is angle bisector of  $\angle ABC$

Diagonals of rhombus and square are perpendicular to each other.



$$\therefore \angle AMB = 90^\circ$$

9. (b); In  $\Delta ABC$ ,

$$AB + BC > AC \quad \dots(i)$$

$$\text{In } \Delta ACD,$$

$$CD + DA > AC \quad \dots(ii)$$

$$\text{In } \Delta BCD,$$

$$BC + CD > BD \quad \dots(ii)'$$

$$\text{In } \Delta ABD$$

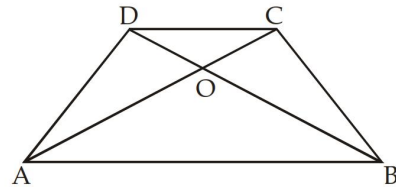
$$DA + AB > BD \quad \dots(iii)$$

Form (i), (ii), (iii) & (iv)

$$2(AB + BC + CD + DA) > 2(AC + BD)$$

$$(AB + BC + CD + DA) > (AC + BD)$$

10. (a);



$$\text{In } \Delta ABC,$$

$$AB > BC$$

$$\therefore \angle ACB > \angle BAC \quad \dots(i)$$

$$\text{In } \Delta ACD,$$

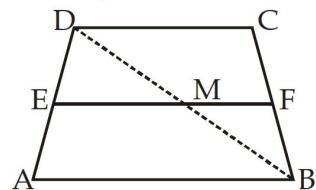
$$\Rightarrow \angle ACD > \angle CAD \quad \dots(ii)$$

$$(\angle ACB + \angle ACD) > (\angle BAC + \angle CAD)$$

$$\angle C > \angle A$$

Similarly,  $\angle D > \angle B$

11. (d); E is mid point of AD and  $EM \parallel AB$ .



M is midpoint of BD.



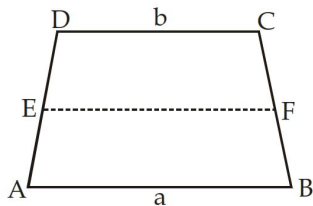
$$EM = \frac{1}{2}AB \quad \dots(i)$$

In  $\triangle BDC$ , M is midpoint of BD and  $MF \parallel DC$ .  
and F is mid point of BC.

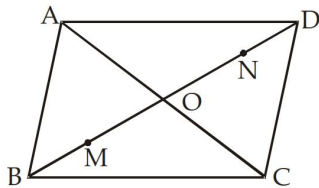
$$MF = \frac{1}{2}DC \quad \dots(ii)$$

$$EF = EM + MF = \frac{1}{2}(AB + DC)$$

12. (b);  $EF = \frac{1}{2}(AB + CD) = \frac{1}{2}(a + b)$



13. (d);  $OB = ON = \frac{36}{2} = 18 \text{ cm}$

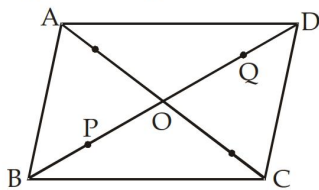


$$OM = \frac{1}{3} \times 18 = 6 \text{ cm}$$

$$ON = \frac{1}{3} \times 18 = 6 \text{ cm}$$

$$MN = 6 + 6 = 12 \text{ cm}$$

14. (b);  $PQ = OP + OQ$



$$OP = \frac{1}{3} \times 9 = 3 \text{ cm}$$

$$OQ = \frac{1}{3} \times 9 = 3 \text{ cm}$$

$$PQ = 3 + 3 = 6 \text{ cm}$$

15. (b); Perimeter of rhombus = 40 cm

$$\text{Side} = \frac{40}{4} = 10 \text{ cm}$$

$$\angle ABD = \angle ADB = 60^\circ [AB = AD]$$

$$\text{Area of rhombus} = 2 \times \frac{\sqrt{3}}{4} \times (AB)^2$$

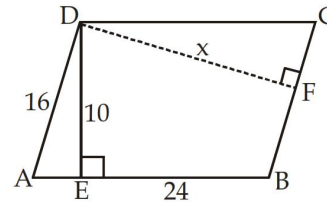
$$= 2 \times \frac{\sqrt{3}}{4} \times 10 \times 10 = 50\sqrt{3} \text{ cm}^2$$

16. (c);  $\angle A + \angle B = 180^\circ$

$$4x + 5x = 180^\circ$$

$$x = 20^\circ, \angle A = \angle C = 20 \times 4 = 80^\circ$$

17. (c);



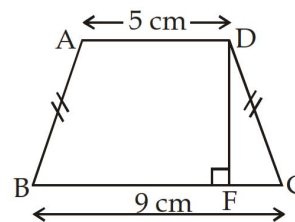
$$\text{Area of parallelogram} = \text{base} \times \text{height}$$

$$= 24 \times 10 = 240 \text{ cm}^2$$

$$\text{Let distance } DF = x \text{ cm}$$

$$16 \times x = 240, \quad x = \frac{240}{16} = 15 \text{ cm}$$

18. (a); Area of trapezium ABCD =  $\frac{1}{2} \times DF \times (AD + BC)$



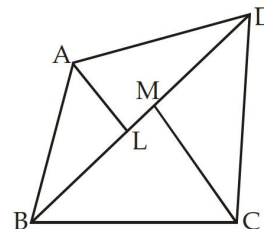
$$35 = \frac{1}{2} \times DF \times (5 + 9)$$

$$DF = 5 \text{ cm}$$

$$FC = \frac{BC - DF}{2} = \frac{9 - 5}{2} = 2 \text{ cm}$$

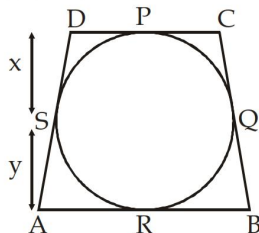
$$CD = \sqrt{(DF)^2 + (FC)^2} = \sqrt{5^2 + 2^2} = \sqrt{29} \text{ cm}$$

19. (d); Area (ABCD) = area ( $\triangle ABD$ ) + area ( $\triangle BCD$ )



$$= \frac{1}{2} \times 64 \times 13.2 + \frac{1}{2} \times 64 \times 16.8 = 960 \text{ cm}^2$$

20. (c);  $DS = PD = x$



$$PC = CQ = 5 - x$$

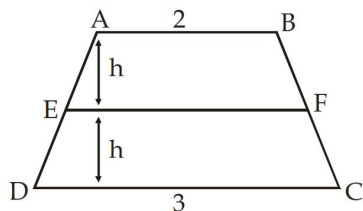
$$BQ = BC - CQ = 7 - (5 - x) = 2 + x$$

$$BQ = BR = 2 + x$$

$$y = 6 - 2 - x = 4 - x$$

$$AD = x + y = 4$$

21. (a);  $\frac{\text{Area of trapezium ABFE}}{\text{Area of trapezium EFCD}}$



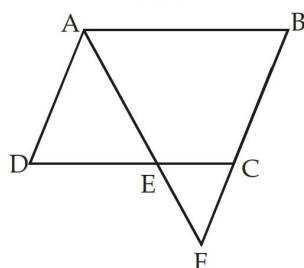
$$= \frac{\frac{1}{2} \left[ 2 + \frac{5}{2} \right] \times h}{\frac{1}{2} \left[ 3 + \frac{5}{2} \right] \times h} \Rightarrow \frac{9}{11}$$

22. (d);  $60^\circ + 120^\circ + 90^\circ + x = 360^\circ$

$$x = 90^\circ = \frac{\pi}{2}$$

23. (d);

24. (a);  $\angle DAE = \angle BAE$



$$\angle BAE = \angle AED$$

$$\therefore \angle DAE = \angle AED$$

$$AD = DE = 6 \text{ cm}$$

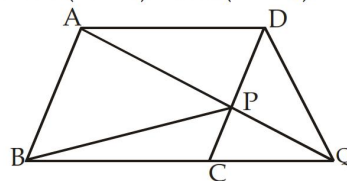
In  $\triangle CEF$ ,

$$\angle EFC = \angle CEF$$

$$CE = CF$$

$$CF = 10 - 6 = 4 \text{ cm}$$

25. (a);  $\text{Area}(\triangle BCP) = \text{Area}(\triangle CPQ)$



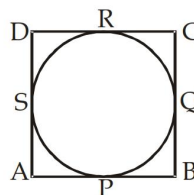
$$\triangle CPQ \cong \triangle ABP$$

$$\text{area}(\triangle CPQ) = \text{area}(\triangle ABP)$$

$$\text{area}(\triangle CPQ) = \text{area}(\triangle DPQ)$$

$$\text{area}(\triangle BCP) = \text{area}(\triangle DPQ)$$

26. (c);



$$AP = AS, BP = BQ,$$

$$CR = CQ \text{ and } DR = DS$$

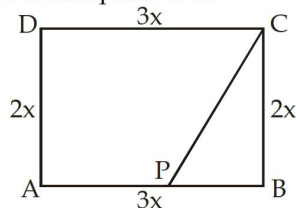
$$AP + BP + CR + DS = AS + BQ + CQ + DS$$

$$(AP + BP) + (CR + DR) = (AS + DS) (BQ + CQ)$$

$$AB + CD = AD - BC$$

$$AD = AB + CD - BC = 6 + 3 - 7.5 = 2.5 \text{ cm}$$

27. (d); P is mid point of AB



$$AP = PB = \frac{3x}{2}$$

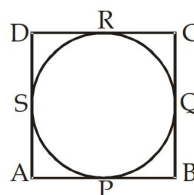
In right  $\triangle CPB$ ,

$$(CP)^2 = (BP)^2 + (BC)^2$$

$$CP = \sqrt{\left(\frac{3x}{2}\right)^2 + (2x)^2} = \frac{5x}{2}$$

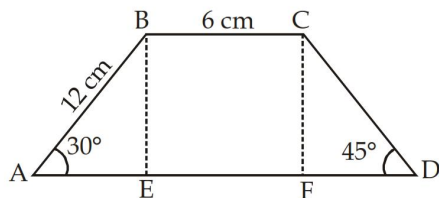
$$\sin \angle CPB = \frac{BC}{CP} = \frac{2x}{\frac{5x}{2}} = \frac{4x}{5x} = \frac{4}{5}$$

28. (b);  $AP = AS, BP = BQ$   
 $CR = CQ, DR = DS$



$$\begin{aligned}\therefore AP + BP + CR + DR &= AS + BQ + CQ + DS \\ (AP + BP) + (CR + DR) &= (AS + DS) + (BQ + CQ) \\ AB + CD &= AD + BC \\ BC &= AB + CD - AD = 6 + 5 - 7 = 4 \text{ cm}\end{aligned}$$

29. (d); Trapezium ABCD



$$\text{In } \triangle ABE, \quad \frac{BE}{AB} = \sin 30^\circ$$

$$\begin{aligned}\therefore BE &= 6 \text{ cm} \\ BE &= CF\end{aligned}$$

$$\cos 30^\circ = \frac{AE}{AB}$$

$$\Rightarrow \frac{\sqrt{3}}{2} = \frac{AE}{12}, \quad AE = 6\sqrt{3} \text{ cm}$$

In  $\triangle CFD$ ,

$$\tan 45^\circ = \frac{CF}{FD}$$

$$\Rightarrow 1 = \frac{6}{FD}$$

$$FD = 6$$

$$\therefore BC = EF = 6 \text{ cm}$$

$$\text{Now area of trapezium} = \frac{1}{2} \times (BC + AD) \times BE$$

$$= \frac{1}{2} (18 + 6\sqrt{3}) \times 6$$

$$= 18(3 + \sqrt{3}) \text{ cm}^2$$

30. (a); Side of square = a

$$\text{Length of diagonal} = a\sqrt{2}$$

$$\frac{\text{Area of sq. of side } a}{\text{Area of sq. side } a\sqrt{2}} = \frac{a^2}{a^2 2} = \frac{1}{2} = 1 : 2$$